CA/Browser Forum Tracker

Part 3 - S/MIME Working Group

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About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. It uses several AI models from competing labs, running on different platforms. One model gathers the current claims on a topic, a different one checks each claim against its source, and disagreements are kept, not smoothed over. The aim is an even-handed account that no outside party has shaped. AI makes mistakes, and so does the system built around it. We find and fix some in every cycle, so the methods keep changing and the content should improve over time. Updates come at least quarterly, sooner if something major happens.

The S/MIME Working Group writes the rules for certificates that sign and encrypt email. Its Baseline Requirements are the youngest of the Forum's documents, adopted at the start of 2023. The current version is 1.0.16, published on 28 September 2026 [1].

This part has four sections. Section 1 covers the requirements and the version history behind them. Section 2 covers the RSA key size transition, whose first deadline arrived on 15 September. Section 3 follows the one ballot that reached a vote this autumn. Section 4 covers the group's post-quantum work, where it remains ahead of the larger server certificate group.

The group's pace picked up after a quiet summer. One ballot was published, one went to a vote, and a post-quantum draft appeared. Every ballot below is given with its stage, because a ballot that has passed its vote is not yet a rule.

1. Requirements and Version History

The version table is the record of what this group has agreed. Reading it in order shows a group that imports the server group's decisions deliberately rather than automatically.

Recent Developments

Version 1.0.16 is the newest change to the email rules. It carries Ballot SMC018, titled Realignment of Multipurpose use cases, and was published on 28 September [1]. The working group chair announced that day that the intellectual property review had ended with no exclusion notices [2]. Publication came four days after the review closed on 24 September [3].

The ballot passed without opposition. It drew 15 yes votes from certificate issuers and three from certificate consumers, with none against and none abstaining [3]. Its voting period ran from 18 to 25 August [3].

Its main effect is a date in 2027. From 1 July 2027, subordinate and subscriber S/MIME certificates must carry the email protection purpose, and may add client authentication but nothing else [1]. The ballot also brings the cross-certificate rules from the CCADB policy into

the S/MIME requirements with immediate effect [3]. It confirms that the legacy certificate profiles were deprecated as of 15 July 2025 [3].

Current State

The versions before it show the same habit of importing server group decisions by explicit ballot. Version 1.0.15 came from Ballot SMC017 on 30 July 2026 and raised the minimum RSA key size for certificate authorities [4]. Version 1.0.14 came from Ballot SMC016 on 5 May 2026 and imported the server group's DNSSEC logging carve-out and SHA-1 sunset [4]. Version 1.0.13 allowed a mobile driving licence to authenticate an individual's identity [5].

Post-quantum algorithms have been permitted here for over a year. Ballot SMC013, titled Introduction of PQC Algorithms, was published on 22 August 2025 as version 1.0.11 [6]. Section 4 describes what has been built on it since.

2. The RSA Key Size Transition

This is the group's most consequential key size change, and its first stage is now in force.

Current State

The first stage took effect on 15 September 2026. A new root or subordinate certificate authority RSA key created from that date must be at least 4096 bits [7]. The version table gives both version 1.0.14 and version 1.0.15 a compliance date of 15 September 2026, and both dates have now been reached [7]. The date passed quietly, with no guidance post on the group's public list or the Forum's site [8].

The increase applies to certificate authority keys only. The minimum for subscriber certificates stays at 2048 bits [4].

The second stage closes the path for legacy keys without forcing their removal. From 15 September 2027, certificate authorities may not issue subscriber certificates from a subordinate authority whose RSA modulus is smaller than 3072 bits [9]. Existing 2048-bit subordinate authorities are not revoked by that rule. They stop being able to issue, which lets them age out as their own certificates expire.

3. The Ballot in Its Vote

One S/MIME ballot reached a vote this autumn, and its result is the next fact this part is waiting on.

Recent Developments

Ballot SMC020 removes the requirement to record which version of the Baseline Requirements a validation method came from [10]. It is modelled on the server group's Ballot SC-099, adapted to S/MIME terms [11]. It also adds audit log requirements for verification activities, effective 15 March 2027 [10]. The log must record the information validated, the domain whose control was validated if it differs from the mailbox domain, and the method used [10].

The ballot moved quickly once it started. It was proposed by the OISTE Foundation and endorsed by DigiCert and Apple [12]. Its discussion period ran from 5 to 12 October 2026, and its voting period from 12 October to 19 October at 19:00 UTC [10].

Current State

The vote has closed and the result is not yet known. As checked on 19 October, no results post had appeared on the group's public list [13]. The readable votes were four in favour from 13 October, from MSC Trustgate, SwissSign, rundQuadrat and Telia Company [12]. The thread had grown to 21 messages, but the public mirror shows only the first eight, so any later votes cannot be read [13].

SMC020 therefore sits between voting and adoption. If it passes, it enters a 30-day intellectual property review before it can be published into a new version.

4. Post-Quantum Email

The S/MIME group admitted post-quantum algorithms a year before any other Forum group, and its next step is the hybrid kind.

Recent Developments

A new draft would add composite ML-DSA to the email rules. Ballot SMC019, titled Introduction of Composite ML-DSA Algorithms, was opened as a pull request on 24 September 2026 by DigiCert [14]. A composite pairs a post-quantum signature with a traditional one in a single certificate. The draft builds on the standalone ML-DSA and ML-KEM support that SMC013 already published [14].

The draft has been narrowed once. A change on 28 September removed the composite pairings with Ed25519 and Ed448 [15]. It has not changed since [16].

A root program has already said it will trust them. Apple announced on 21 September that it will accept composite roots that pair ML-DSA with ECDSA or RSA for S/MIME email [17]. Part 4 covers the rest of that announcement, which concerns web certificates.

Current State

SMC019 is a draft and has not entered any stage of the ballot lifecycle. No discussion period has been announced, and a search of the group's public list for the ballot returns nothing [16]. It does not appear on the group's ballots page [18].

The draft also waits on the IETF. It cites Internet-Drafts for the composite algorithms, with a note to update the references once RFC numbers are assigned [14]. The composite signatures draft for X.509 reached final review at the RFC Production Center on 14 October 2026 [19]. It had no RFC number as of 18 October [19].

Unresolved Conflicts

Two questions are open here. The first is why SMC019 is missing from the group's ballots page, while SMC020 was listed there as a draft before its discussion began [18]. The record does not say. An announced discussion period, or a ballots page that lists it, would settle the status.

The second concerns what SMC013 permits. The September edition could not say which post-quantum algorithms it allows. The SMC019 draft states that they are standalone ML-DSA and ML-KEM, in two numbered key subsections of the certificate profile [14]. That is the draft's description of the published text, not a reading of the ballot itself. Reading SMC013 would confirm the conditions attached to each algorithm.

Sources

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